With $r = n$, (8) gives, for every $x(0)$,

$$x(\infty) = Vu, \tag{9}$$

with V as in Corollary 3.3.

The condition of Corollary 3.3 can also be read on the matrix ΛW with the tools of the course. The matrix ΛW is no longer row-stochastic: since W is row-stochastic, the i-th row of ΛW sums to λ_i. Rows of agents that are not prejudiced sum to 1, rows of prejudiced agents sum to less than 1, and since $\Lambda \neq I$ at least one such row exists. Hence ΛW is row-substochastic (Definition 1.15) and Lemma 1.16 applies.

For the matrix ΛW, the weighted out-degree of agent i is λ_i: the nodes with out-degree 1 are the agents that are not prejudiced, and the nodes with out-degree less than 1 are the prejudiced agents. The condition of Lemma 1.16 therefore reads: every agent that is not prejudiced is influenced, directly or through intermediaries, by a prejudiced agent, i.e. all the agents are P-dependent. This is exactly the condition of Corollary 3.3. Indeed, [5] remarks that Corollary 3.3 can be turned into a criterion of Schur stability for substochastic matrices, and [4] proves the same condition for the FJ model through this lemma.

Two observations of ours give an intuition for Corollary 3.3. First, letting $k \to +\infty$ in (4) shows that the limit $x(\infty) = Vu$ is an equilibrium, $x(\infty) = \Lambda W x(\infty) + (I - \Lambda)u$. Letting $e(k) = x(k) - x(\infty)$ and subtracting this equation from (4) gives

$$e(k+1) = \Lambda W\, e(k), \tag{10}$$

an autonomous linear system in which the prejudice u does not appear: ΛW decides whether the opinions converge, u decides where.

Second, the linear system (10) shows the role of the P-dependent agents. At each step, agent i keeps a fraction λ_i of the weighted average of the deviations of the agents it listens to. An agent that is not prejudiced ($\lambda_i = 1$) only averages the deviations, while a prejudiced agent loses a fraction $1 - \lambda_i$ of them (in (4), this fraction is replaced by the agent's fixed prejudice). The deviations therefore leave the group only through the prejudiced agents, and they vanish only if every agent is influenced, directly or through intermediaries, by one of them, i.e. all agents are P-dependent. A group of agents that is not influenced, even indirectly, by any prejudiced agent loses no deviation and behaves as an FDG model, as in (7).

In social terms, the group converges, for every initial opinion, as soon as every agent that is not anchored to a prejudice listens, directly or through intermediaries, to someone who is. On a strongly connected network, such as the two networks of Figure 1, a single prejudiced agent is enough. Note that this does not require aperiodicity, unlike the FDG model of Section 2.2.

If some agents are P-independent, the model is not asymptotically stable, but it may still converge. Theorem 3.2 gives the condition in terms of W^{22}; in terms of the whole matrix ΛW it reads as follows.

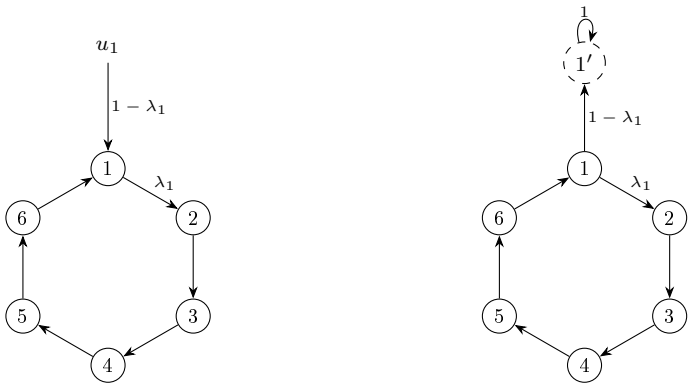
Corollary 3.4 ([5]). *The model* (4) *converges if and only if* ΛW *is regular, i.e. the limit* $\lim_{k\to\infty}(\Lambda W)^k$ *exists.*

As remarked in [5], this property is not trivial: a system $x(k+1) = Ax(k) + Bu$ with a regular matrix A may not converge, and may even have unbounded solutions, as for $A = B = I$.

The decomposition (6)–(7) also explains these results on the spectrum (this remark is ours). Listing the P-dependent agents first, ΛW is block triangular,

$$\Lambda W = \begin{bmatrix} \Lambda^{11} W^{11} & \Lambda^{11} W^{12} \\ 0 & W^{22} \end{bmatrix}, \qquad \operatorname{spec}(\Lambda W) = \operatorname{spec}(\Lambda^{11} W^{11}) \cup \operatorname{spec}(W^{22}),$$

and $\rho(\Lambda^{11} W^{11}) < 1$ by Theorem 3.2. Hence the eigenvalues on the unit circle, which decide convergence and asymptotic stability (Theorem 1.9), can only come from W^{22}, i.e. from the FDG model of the P-independent agents: the model is asymptotically stable when there are no P-independent agents ($r = n$), and it converges when the eigenvalue 1 of W^{22} is semi-simple and its other eigenvalues lie inside the unit disk, i.e. when W^{22} is regular.



(a) FJ model on the directed cycle (b) augmented graph

Figure 4: The FJ model on the directed cycle of Figure 1(a) when agent 1 is the only prejudiced agent ($\lambda_1 < 1$, $\lambda_i = 1$ for $i \neq 1$), and its augmented graph. Unlabelled edges have weight 1. (a) The prejudice u_1 enters agent 1 as an external input, with weight $1 - \lambda_1$; this arrow is not an edge of G. (b) The input becomes the virtual agent $1'$ (dashed), which is stubborn and keeps the opinion u_1. The virtual agents $2', \dots, 6'$ are stubborn agents that nobody listens to, and are not drawn.

The augmented graph also shows, from the side of the FDG model, why consensus is not the right notion for the FJ model. This observation is ours. The augmented model (11) is an FDG model, so the classification of Table 1 applies to it. Each virtual agent is influenced only by itself, so it is a sink of the condensation digraph of the augmented graph, and it is aperiodic because of its self-loop. Hence the augmented graph has at least n sinks and it is not strongly connected; for $n \geq 2$, the consensus cases (i) and (iii) of Table 1 never occur, and neither does case (ii). Only two outcomes remain.

- *Disagreement* (case (iv)), when all the sinks are aperiodic. If all agents are P-dependent, the only sinks are the n virtual agents, since every real agent is influenced, directly or through intermediaries, by some prejudiced agent j, and hence by j'. This is the situation of Remark 2.5: by Theorem 2.3, every real agent converges to a convex combination of the opinions of the stubborn virtual agents, i.e. of the prejudices, which is exactly (9). If some agents are P-independent, the sinks of C^{22} are further sinks of the augmented graph, and all of them are aperiodic if and only if W^{22} is regular.

- *No convergence* (case (v)), when at least one sink of C^{22} is periodic, i.e. when W^{22} is not regular.

These two outcomes are exactly the two alternatives of Theorem 3.2. The disagreement of case (iv) concerns all the $2n$ agents of the augmented model, whose virtual agents keep their different values u_i forever ($x'(0) = u$ is part of the initial condition); it does not imply that the final opinions of the real agents differ, which, as discussed above, depends on V and u.

Two descriptions of the same dynamics. When all the agents are P-dependent, the FJ model is asymptotically stable (Corollary 3.3), while the augmented model (11), whose matrix is row-stochastic and has the eigenvalue 1 (Theorem 1.7), is never asymptotically stable: it converges to the disagreement of case (iv), which depends on the initial opinions $x'(0) = u$ of the virtual agents. The two statements describe the same trajectories and do not contradict each other (this reading is ours). ~~The~~ Let $\bar{A}$ be the matrix of (11). It is block triangular, ~~so its eigenvalues are those of ΛW,~~ and so is ΛW when the P-dependent agents are listed first (Section 3.1.2); hence, in general,

$$\operatorname{spec}(\bar{A}) = \underbrace{\operatorname{spec}(\Lambda^{11} W^{11})}_{\text{P-dependent agents}} \cup \underbrace{\operatorname{spec}(W^{22})}_{\text{P-independent agents}} \cup \underbrace{\{1\}_n}_{\text{virtual agents}}, \tag{12}$$

where $\{\mu\}_m$ denotes the eigenvalue μ repeated m times. Each piece belongs to a group of states of the augmented model: the P-dependent real agents, with eigenvalues inside the unit disk ~~, and the~~ (Theorem 3.2), whose perturbations are forgotten; the P-independent real agents, which follow the

FDG model with matrix W^{22} and bring its eigenvalue 1 and, if W^{22} is not regular, other eigenvalues on the unit circle; the n virtual agents, whose block is the identity and whose eigenvalue 1 ~~with~~ keeps each prejudice. The FJ model, in which u is an input, has only the first two pieces. Hence the FJ model is asymptotically stable if and only if the second piece is missing ($r = n$, Corollary 3.3), the two models converge if and only if W^{22} is regular (Theorem 3.2 and the two outcomes above), and the third piece makes the augmented model never asymptotically stable. When all the agents are P-dependent, the eigenvalue 1 has multiplicity n, semi-simple by Theorem 2.3~~applied to the n aperiodic sinks~~, and by (11) its eigenvectors are

$$\bar{A}\begin{bmatrix} a \\ z \end{bmatrix} = \begin{bmatrix} a \\ z \end{bmatrix} \iff a = (I - \Lambda W)^{-1}(I - \Lambda)\, z = Vz, \qquad z \in \mathbb{R}^n :$$

one direction for each virtual agent, and the limit (Vu, u) of the augmented ~~graph~~model is the eigenvector with $z = u$: the matrix V of the FJ model describes the eigenspace of the eigenvalue 1 of the augmented model. The difference lies in what is a state. In the FJ model u is an input: asymptotic stability concerns the perturbations of $x(0)$ with u fixed, which die out as in (10), while changing u is not a perturbation but a different input, with a different equilibrium Vu. In the augmented model u is part of the state: changing it is a perturbation of the initial condition along the directions of the eigenvalue 1, and it is never forgotten; perturbing only the real agents, with $x'(0)$ unchanged, the augmented model forgets the perturbation exactly as the FJ model. ~~The rewriting of Section 3.2.1 shows the same fact in the opposite direction: the opinion of a stubborn agent is a state in the FDG model, which is not asymptotically stable, andan input in the FJ model, which is~~Sections 3.2.1 and 3.2.2 read the examples of Figures 7 and 12 in these terms.

3.1.4 Influence centrality

In the FDG model the social power w is defined only when the group reaches a consensus (Section 2.1.2), which is not the relevant notion for the FJ model (Section 3.1.3). Nor does the augmented FDG model (11) help: its graph has at least n sinks, so for $n \geq 2$ it never reaches a consensus and its social power is not defined (this remark is ours). A different measure is needed, and [5] discusses an extension of the social power to the FJ model, introduced by Friedkin.

Influence centrality. As in [5], we confine ourselves to the asymptotically stable FJ model with $u = x(0)$, the traditional assumption of the Friedkin–Johnsen theory (Section 3.1.1); then $x(\infty) = Vx(0)$ by (9). The assumption $u = x(0)$ is what gives the initial opinions a role: in general $x(\infty) = Vu$ does not depend on $x(0)$, so the initial opinions would have no weight in the final opinions. The social power of agent i is generalized as the mean weight of its initial opinion in determining the group members' final opinions [5]. These mean weights are the elements of the non-negative vector c such that

$$c = n^{-1}V^\top \mathbf{1}_n, \qquad \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i(\infty) = \frac{1}{n}\mathbf{1}_n^\top V x(0) = c^\top x(0), \tag{13}$$

and c_i is called the *influence centrality* of agent i [5]. In terms of V, c_i is the average of the i-th column of V, whose entries are the weights of $x_i(0)$ in the final opinions of the single agents. Since V is stochastic, c is non-negative and $c^\top \mathbf{1}_n = 1$, like w (Theorem 2.1), so the two measures can be compared directly. Similar to French's social power, the influence centrality is generated by an opinion formation mechanism [5].

The relation with the social power is made precise in [5]. The convergence and the consensus of the FDG model are equivalent, respectively, to the regularity and the full regularity of the stochastic matrix A [5] (Definition 1.10). For a fixed W, let $\Lambda = \alpha I_n$ with $\alpha \in (0, 1)$, i.e. the same susceptibility for all agents, and consider the corresponding matrix V_α and vector c_α. For $\alpha = 0$, i.e. $\Lambda = 0$, every agent keeps its initial opinion, $V = I$ and $c_0 = n^{-1}\mathbf{1}_n$: the social power is uniformly distributed between all agents [5].

Lemma 3.5 ([5]). *Let W be a fully regular matrix and w stand for the vector of French's social power, corresponding to the FDG model* (1) *with $A = W$. Then, $w = \lim_{\alpha \to 1-0} c_\alpha$.*

Under the assumption of Corollary 3.6, every agent of the FJ model obtained by the rewriting is prejudiced or influenced by a prejudiced agent, so the FJ model is asymptotically stable (Corollary 3.3), whereas the FDG model is convergent but not asymptotically stable, since a row-stochastic matrix always has the eigenvalue 1 (Theorem 1.7, see also [5]). The explanation given in [5] is that changing the initial condition $x(0)$ of the FJ model leaves the prejudice vector u unchanged, while in the FDG model the prejudice is part of the state vector.

The spectra make this explanation precise, in the direction FJ $\to$ augmented $\to$ FDG (this computation is ours). Let S be the set of the s stubborn agents and R the set of the other $n-s$ agents, and list the agents of S first. The rows of A are split accordingly: a stubborn agent has only its self-loop, so its row is $e_i^\top$; for the agents of R, the block A^{RR} contains the weights they give to each other and the block A^{RS} the weights they give to the stubborn agents. In the FJ model the agents of R have $\lambda_i = 1$ and the same rows, while the rows of S are multiplied by $\lambda_i = 0$:

$$\Lambda W = \begin{bmatrix} 0 & 0 \\ A^{RS} & A^{RR} \end{bmatrix}, \qquad A = \begin{bmatrix} I_s & 0 \\ A^{RS} & A^{RR} \end{bmatrix}.$$

FJ model. ΛW is block triangular, so $\operatorname{spec}(\Lambda W) = \{0\}_s \cup \operatorname{spec}(A^{RR})$, and asymptotic stability (Corollary 3.3) gives $\rho(A^{RR}) < 1$.

Augmented model. Listing the virtual agents in the same order, first S' and then R', and using $1-\lambda_i = 1$ on S and $1-\lambda_i = 0$ on R, the matrix of (11) is

$$\bar{A} = \left[\begin{array}{cc|cc} 0 & 0 & I_s & 0 \\ A^{RS} & A^{RR} & 0 & 0 \\ \hline 0 & 0 & I_s & 0 \\ 0 & 0 & 0 & I_{n-s} \end{array}\right] \begin{matrix} S \\ R \\ S' \\ R' \end{matrix}$$

Its spectrum is that of ΛW plus one eigenvalue 1 for each virtual agent, as in (12), but the virtual agents play two different roles. The rows and columns of R' are zero outside the block I_{n-s}: nobody listens to the virtual agents of the non-stubborn agents, and their $n-s$ eigenvalues 1 are decoupled from the real agents. The virtual agent i' of a stubborn agent is listened to only by agent i, with weight 1, so $x_i(k) = x_{i'}(k-1) = u_i$ for $k \geq 1$.

FDG model. Removing R' and merging each stubborn agent i with its virtual agent i' gives exactly A: the merged agent keeps the self-loop of i' and is listened to by R with the weights A^{RS} of i. Hence $\operatorname{spec}(A) = \{1\}_s \cup \operatorname{spec}(A^{RR})$: the eigenvalues 0 of the stubborn rows of ΛW are replaced by the eigenvalues 1 of their virtual agents. The eigenvalue 1 that forbids the asymptotic stability of the FDG model is that of a virtual agent: the stubborn opinion is an input in the FJ model and the state of a virtual agent in the FDG model.

Figure 7 starts the two models again from $x_1(0) = 0.5$, leaving u unchanged: now $u_1 = 0 \neq x_1(0)$, and agent 1 of the FJ model is prejudiced but no longer stubborn.

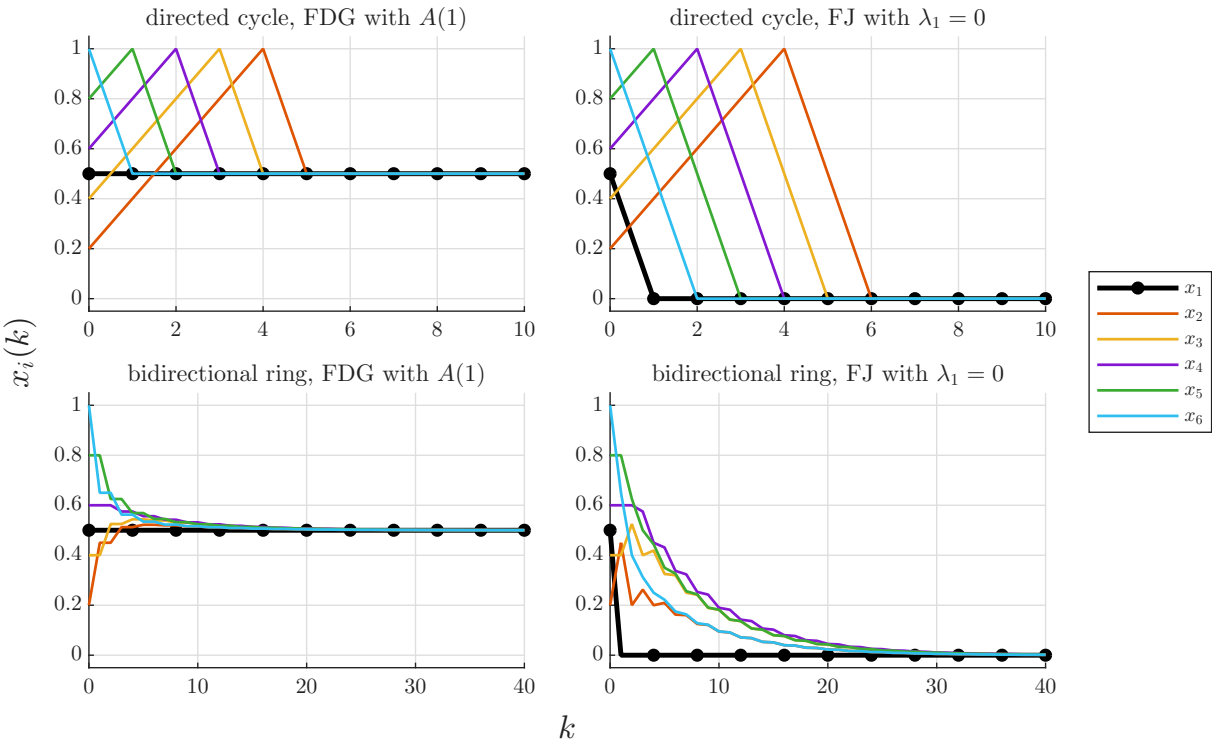


Figure 7: The models of Figure 6 started from $x_1(0) = 0.5$, with the other initial opinions and the prejudice vector $u = (0,\ 0.2,\ 0.4,\ 0.6,\ 0.8,\ 1)^\top$ unchanged, so $u_1 \neq x_1(0)$. Left: FDG with $A(1)$; right: FJ with $\lambda_1 = 0$. Top: directed cycle, $K = 10$; bottom: bidirectional ring, $K = 40$.

Figure 7 shows this difference. Moving $x_1(0)$ from 0 to 0.5 moves the stubborn agent of the FDG model, which keeps the new value forever, and the whole group follows it to 0.5. In the FJ model the same change does not touch $u_1 = 0$: agent 1 goes back to u_1 at the first step, and the group converges to the same limit Vu as before. In social terms, in the FDG model stubbornness is a memory inside the state, which is never forgotten; in the FJ model it is an external input, and the state forgets $x(0)$.

How the perturbation evolves. Figure 7 perturbs the initial state. Let $x(k)$ and $\tilde{x}(k)$ be the trajectories of Figures 6 and 7~~, started from $x_1(0) = 0$ and $x_1(0) = 0.5$. Figure ?? shows, for each agent, their difference $\delta(k) = \tilde{x}(k) - x(k)$, with $\delta(0) = 0.5\, e_1$.~~

~~The difference $\delta(k)$ shows where the perturbation of $x_1(0)$ is at each step~~. In the FJ model ~~, subtracting the two equations (4) written for $\tilde{x}$ and for x, the term $(I - \Lambda)u$ cancels, since the prejudices are the same , and~~ the prejudice u is the same in the two runs, so their difference obeys (this computation is ours)

$$\delta(k+1) = \Lambda W\, \delta(k).$$

$$\delta(k+1) = \Lambda W\, \delta(k),$$

~~This is~~ the same recursion as ~~(10): both $\delta(k)$ and $e(k)$ are differences of two solutions with the same u, the second one being, for $e(k)$, the constant solution $x(\infty)$. The two quantities, however, measure different things:~~ the error (10), although $e(k)$ ~~tells how far a single trajectory still is~~ measures the distance of one trajectory from its limit ~~,~~ and $\delta(k)$ ~~how much~~ what is left of the perturbation~~is still present.~~

~~Difference $\delta_i(k) = \tilde{x}_i(k) - x_i(k)$ between the trajectories of Figure 7 and those of Figure 6, i.e. the effect of changing $x_1(0)$ from 0 to 0.5. Top: FDG with $A(1)$; bottom: FJ with $\lambda_1 = 0$. Left: directed cycle, $K = 10$; right: bidirectional ring, $K = 40$. The thin grey line marks $\delta_i = 0$.~~

. Since all agents are P-dependent, ΛW is Schur stable (Corollary 3.3)~~and $\delta(k) \to 0$:~~ : $\delta(k) \to 0$ and the two runs reach the same limit Vu, which is what asymptotic stability means. In the FDG model the same computation gives $\delta(k+1) = A(1)\,\delta(k)$, and the perturbation is ~~forgotten~~never forgotten.

The spectra of the example locate the perturbation (this computation is ours). Here $S = \{1\}$ and A^{RR} is the matrix of agents $2, \ldots, 6$. On the directed cycle~~it travels backwards along the cycle, from ,~~

removing agent 1 ~~to agents 6, 5, . . . , 2, and when it comes back to~~ breaks the cycle: agents $2, \dots, 6$ form a chain without closed walks, $(A^{RR})^5 = 0$ and all the eigenvalues of A^{RR} are 0. On the bidirectional ring they form a path with weights $\frac{1}{2}$, and the eigenvalues of A^{RR} are $0, \pm\frac{1}{2}, \pm\frac{\sqrt{3}}{2}$ (computed numerically). Hence

	directed cycle	bidirectional ring
$\mathrm{spec}(\Lambda W)$	$\{0\}_6$	$\{0\}_2,\ \pm\frac{1}{2},\ \pm\frac{\sqrt{3}}{2}$
$\mathrm{spec}(A(1))$	$1,\ \{0\}_5$	$1,\ 0,\ \pm\frac{1}{2},\ \pm\frac{\sqrt{3}}{2}$

The two matrices differ only in the row of agent 1 ~~it is cancelled, since $\lambda_1 = 0$: $\delta(k) = 0$ for $k \geq 6$. On the bidirectional ring it is split between the neighbours and fades gradually.~~, $\Lambda W = A(1) - e_1 e_1^\top$, and so do the two spectra: the eigenvalue of agent 1 is 0 in ΛW and 1 in $A(1)$. The perturbation $\delta(0) = 0.5\, e_1$ falls exactly on agent 1.

- In the FDG model ~~there is no input to cancel, and the same computation gives $\delta(k+1) = A\,\delta(k)$ (ours). Agent~~, $w = e_1$ is the left eigenvector of the eigenvalue 1, so $w^\top \delta(k+1) = w^\top A(1)\,\delta(k) = w^\top \delta(k)$: the quantity $\delta_1(k) = 0.5$ is conserved, and by Theorem 2.1 $\delta(k) \to 0.5\,\mathbf{1}_6$.
- In the FJ model agent 1 discards the perturbation at the first step, $\delta_1(1) = 0$, and what has reached its neighbours evolves with A^{RR}, inside the unit disk; on the directed cycle $\delta(k) = 0$ for $k \geq 6$.
- In the augmented model, the eigenvalue 1 that ΛW lacks is that of the virtual agent $1'$. Figure 7 moves $x_1(0)$ and leaves $x_{1'}(0) = u_1$ unchanged, so it does not touch this eigenvalue; moving u_1 would reproduce the FDG panel.

The two panels of Figure 7 therefore differ not in the dynamics but in what is perturbed: the opinion of agent 1 is the ~~only globally reachable node, with social power $w = e_1$, so by Theorem 2.1 $\delta(k) \to (w^\top \delta(0))\mathbf{1}_6 = 0.5\,\mathbf{1}_6$: the stubborn agent~~keeps the perturbation and passes it to everybody. state of the virtual agent $1'$ in the FDG model, and only the initial opinion of a real agent in the FJ model.

The stubborn agent as a prejudice of its neighbours. The rewriting of [5] keeps the stubborn agent as an agent of the FJ model. A different rewriting, which is ours, removes it. Consider the bidirectional ring with $t = 1$ (Figure 8(a)): agent 1 is stubborn, and agents 2 and 6 listen to it with weight 1/2. Since $x_1(k) = x_1(0)$ for all k, the update of agent 6 is

$$x_6(k+1) = \tfrac{1}{2}\,x_5(k) + \tfrac{1}{2}\,x_1(k) = \tfrac{1}{2}\,x_5(k) + \tfrac{1}{2}\,x_1(0),$$

which is the update (4) of an agent with $\lambda_6 = 1/2$, prejudice $u_6 = x_1(0)$ and influence weight 1 on agent 5; the same holds for agent 2. Hence agents $2, \dots, 6$ obey an FJ model on five agents (Figure 8(b)), with $\lambda_2 = \lambda_6 = 1/2$, $u_2 = u_6 = x_1(0)$ and $\lambda_3 = \lambda_4 = \lambda_5 = 1$, whose trajectories coincide with those of the FDG model (bottom-left panel of Figure 6): the stubborn agent has become the common prejudice of the agents that listen to it. Conversely, the augmented graph of this FJ model (Section 3.1.3) has two virtual agents $2'$ and $6'$, both with opinion $x_1(0)$; merging them into a single node gives back the FDG model, in which the stubborn agent 1 plays the role of a virtual agent.

Panel (a) is the situation of Figure 9 again: agent 2 is in the closed group, every agent is connected by a walk to it ($r = n$), the model is asymptotically stable (Corollary 3.3) and $x(\infty) = u_2\mathbf{1}_6 = 0.5\,\mathbf{1}_6$, although the closed group is periodic.

In panels (b)–(d) agent 5 is outside the closed group, and agents 1, 2, 3 and 6 are P-independent: by (7) they obey the FDG model with matrix W^{22}, as if the prejudice did not exist, and their trajectories in Figure 12 are those of the FDG model on W_p and W_a. Renumbering the agents as in (6)–(7), i.e. in the order $\sigma = (4, 5 \mid 1, 2, 3, 6)$, means applying the permutation matrix $P = [\,e_4\ e_5\ e_1\ e_2\ e_3\ e_6\,]^\top$, so that agent i of the new numbering is agent $\sigma(i)$ of the original one; for W_a,

$$\tilde{W}_\mathrm{a} = P\,W_\mathrm{a}P^\top = \left[\begin{array}{cc|cccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ \hline 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{array}\right] = \begin{bmatrix} W^{11} & W^{12} \\ 0 & W^{22} \end{bmatrix}, \qquad \tilde{\Lambda} = P\Lambda P^\top = \mathrm{diag}(1,\, 0.5 \mid 1,\, 1,\, 1,\, 1),$$

$$W^{11} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad W^{12} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad W^{22} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}.$$

The lower-left block is zero because no P-independent agent listens to a P-dependent one; with W_p the second row of W^{22} would be $(0, 0, 1, 0)$. By Theorem 3.2 the FJ model converges if and only if W^{22} is regular, and then the P-dependent agents follow (8), with $\Lambda^{11} = \mathrm{diag}(1,\, 0.5)$ and

$$V = (I_2 - \Lambda^{11}W^{11})^{-1}\big[\,I_2 - \Lambda^{11},\ \Lambda^{11}W^{12}\,\big] = \left[\begin{array}{cc|cccc} 0 & 0.5 & 0 & 0 & 0 & 0.5 \\ 0 & 0.5 & 0 & 0 & 0 & 0.5 \end{array}\right],$$

i.e. $x_4(\infty) = x_5(\infty) = 0.5\,u_5 + 0.5\,x_6(\infty)$.

- In (b) the sink $\{1, 2, 3\}$ of W^{22} is periodic, W^{22} is not regular and the model does not converge: agents $1, 2, 3, 6$ oscillate, and agents $4, 5$, which listen to them, oscillate too.
- In (c) the edge $2 \to 1$ makes the sink aperiodic, W^{22} is regular and the model converges (Corollary 3.4): agents $1, 2, 3$ reach the FDG consensus $0.4\,x_1(0) + 0.4\,x_2(0) + 0.2\,x_3(0) = 0.4$, weighted by the social power of the closed group, agent 6 follows agent 1, and agents $4, 5$ settle at $0.5 \cdot 0.9 + 0.5 \cdot 0.4 = 0.65$.
- The model of (c) is not asymptotically stable, since not all the agents are P-dependent (Corollary 3.3): every consensus vector is an equilibrium of (7), and the initial opinions of the closed group decide which one is reached. In (d), moving only $x_2(0)$ from 0.5 to 1 moves the consensus of the closed group to 0.6 and the P-dependent agents to 0.75, unlike in Figure 7, where the asymptotically stable FJ model forgets the change of $x_1(0)$.

The spectrum of the augmented model, split as in (12), adds two things to this picture (this computation is ours; (d) has the matrices of (c)).

Panel	$\mathrm{spec}(\Lambda^{11}W^{11})$	$\mathrm{spec}(W^{22})$	eigenvalues of $\bar{A}$ on the unit circle
(a)	$2^{-1/3}\{1, e^{\pm 2\pi i/3}\},\ \{0\}_3$	–	$\{1\}_6$
(b)	$\{0\}_2$	$1,\ e^{\pm 2\pi i/3},\ 0$	$\{1\}_7,\ e^{\pm 2\pi i/3}$
(c), (d)	$\{0\}_2$	$1,\ \frac{-1\pm i}{2},\ 0$	$\{1\}_7$

First, it shows why the periodicity of the closed group plays no role in (a). On a cycle of three agents the characteristic polynomial is μ^3 minus the product of the weights along the cycle. In W_p the product is 1, and the eigenvalues $1, e^{\pm 2\pi i/3}$ are the roots of $\mu^3 = 1$; in ΛW the prejudice of agent 2 multiplies its weight by $\lambda_2 = \frac{1}{2}$, and they become the roots of $\mu^3 = \frac{1}{2}$, of modulus $2^{-1/3} \approx 0.79 < 1$. In (b), where the prejudice does not reach the cycle, they stay on the unit circle.

Second, it shows what the model of (c) remembers. Besides the six eigenvalues 1 of the virtual agents, $\bar{A}$ has a seventh one, from W^{22}, whose left eigenvector is the social power $y = (0.4, 0.4, 0.2)$ of the closed group on agents $1, 2, 3$, and zero elsewhere. As for w in Figure 7, the quantity $y^\top x(k) = 0.4\,x_1(k) + 0.4\,x_2(k) + 0.2\,x_3(k)$ is conserved, like the opinion of a virtual agent. In (d) the change of $x_2(0)$ by 0.5 changes it by $0.4 \cdot 0.5 = 0.2$, which is exactly the shift of the closed group from 0.4 to 0.6. The closed group is a memory inside the real agents: this is why the FJ model of (c) converges without being asymptotically stable.

In social terms, a prejudice acts only on the agents that listen, directly or through intermediaries, to the prejudiced agent: when it reaches everybody, the FJ model is asymptotically stable and the structure of the network does not matter; when it does not, the agents it does not reach form an FDG model of their own, which needs aperiodicity to converge and remembers their initial opinions. Table 2 summarizes the four cases.

Panel	W	Prejudiced agent	P-dependent	Outcome	What it shows
(a)	W_p	2, inside the closed group	all ($r = n$)	asymptotically stable (Corollary 3.3), $x(\infty) = u_2 \mathbf{1}_6 = 0.5\,\mathbf{1}_6$	the periodic cycle plays no role
(b)	W_p	5, outside the closed group	$4, 5$ ($r = 2$)	W^{22} not regular: not convergent (Theorem 3.2)	aperiodicity is needed again, for the P-independent agents
(c)	W_a	5, outside the closed group	$4, 5$ ($r = 2$)	W^{22} regular: convergent (Theorem 3.2), not asymptotically stable; agents $1, 2, 3, 6$ at 0.4, agents $4, 5$ at 0.65	convergence without asymptotic stability (Corollary 3.4)
(d)	W_a	as (c), $x_2(0) = 1$, u unchanged	$4, 5$ ($r = 2$)	as (c), but agents $1, 2, 3, 6$ at 0.6, agents $4, 5$ at 0.75	the limit depends on the initial opinions of the P-independent agents

Table 2: Summary of Figure 12: a single prejudiced agent with $\lambda = 0.5$ on the two versions of the auxiliary network (Figure 11).

3.2.3 Prejudices, stubbornness and influence centrality

The assumptions behind the influence centrality. The influence centrality generalizes French's social power [5]: w_i is the share of agent i in the opinion the group agrees on, while $c_i = (n^{-1} V^\top \mathbf{1}_n)_i$, the average of column i of V, tells how much the initial opinion of agent i weighs, on average, in final opinions that may differ. The assumptions, however, are of a different kind. The social power only needs what makes it meaningful, a consensus, $A^\infty = \mathbf{1}_n w^\top$. The influence centrality needs two further assumptions, which the FJ model does not need in order to make sense: $\rho(\Lambda W) < 1$, i.e. every agent is reached by some prejudice, so that V exists and is stochastic; and $u = x(0)$, traditional in the theory of Friedkin and Johnsen [5], without which $x(\infty) = Vu$ would not depend on $x(0)$ and the initial opinions would weigh nothing (this distinction is ours). In return, with $\Lambda = \alpha I$ every agent is a little anchored and $\rho(\alpha W) = \alpha < 1$ for every stochastic W: c_α exists on every network, also where w does not, and PageRank is one of these centralities [5]. Since w weighs the initial state of a model that remembers $x(0)$, and c the input of a model that forgets it, we do not compare them directly, but in parallel.

A single anchored agent. With agent 1 as the only prejudiced agent (Figure 9), $V = \mathbf{1}_6 e_1^\top$ gives $c = e_1$ for every $\lambda_1 < 1$: agent 1, however loosely anchored, collects all the influence centrality, whereas in the FDG model its social power grows gradually with its self-weight, from 1/6 to 1 (Figure 3). Inertia earns influence little by little, an anchor takes it all at once.